Technology & Research

Advancing Human Interface Systems Through Wearable Innovation

Thorbok Tech is a research and innovation entity focused on the development of advanced wearable interface systems, assistive technologies, and human-centered design solutions.

The company explores the intersection of smart textiles, accessibility, and interactive systems, with the goal of enhancing human capabilities, safety, and communication in real-world and extreme environments.

CORE RESEARCH AREAS

- Human-Computer Interaction (HCI)
- Wearable Interface Systems
- Smart Textiles & Responsive Materials
- Assistive Technology & Accessibility
- Human-Centered Design
- Future Human Interface Systems

Flagship Research Project

Luminous Braille Wearable Interface System

This research project introduces a novel wearable interface system integrating luminous braille technology into textile structures.

The system combines tactile braille interaction with light-emitting components, enabling both physical reading and visual signaling.

This dual-modality approach enhances accessibility, improves safety in low-visibility environments, and introduces new possibilities for human-to-human and human-environment communication.

Potential applications include assistive technology, smart wearable systems, emergency identification, and advanced environments such as aerospace and space exploration.

Publications & Research

• arXiv (submission in progress)
• GitHub Repository
https://github.com/ThorbokTech/madison-color-luminous-braille

• Medium Article (public overview)

Scientific Approach

Thorbok Tech develops conceptual and forward-looking research frameworks based on:

• Human-centered design principles
• Interdisciplinary innovation
• Scalable wearable system architecture
• Real-world accessibility challenges

The objective is to bridge the gap between conceptual research and future applied technologies.

Future Research Directions

• Integration of responsive and intelligent textile materials
• Development of adaptive wearable communication systems
• Exploration of human-interface systems in extreme environments
• Applications in aerospace and space-related human systems

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About the Founder

Mason Ewing is a designer, entrepreneur, and independent researcher specializing in wearable interface systems and inclusive innovation.

As a blind creator, his work is directly informed by real-world accessibility challenges, shaping a unique approach to human-centered technology design.